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Prediction and control of crystal shape distribution in the presence of crystal growth modifiers

Aniruddha Majumder [a], Zoltan K. Nagy [a,b,*]

[a] Department of Chemical Engineering, Loughborough University, Loughborough LE11 3TU, UK
[b] School of Chemical Engineering, Purdue University, West Lafayette, IN 47907-2100, USA

HIGHLIGHTS

- Effect of growth modifiers on crystal growth is modelled using the pining mechanism.
- Bivariate population balance model is used to predict crystal shape distribution.
- A population balance model combined with crystal impurity model is proposed.
- A hybrid batch-continuous crystallisation setup is used for feedback control of shape.
- Mean aspect ratio is controlled by manipulating growth modifier concentration.

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ABSTRACT

The crystal product quality, functionality and properties such as flow characteristics are greatly influenced by the crystal size and shape distributions. Hence, prediction and control of crystal size and shape distributions are important from the viewpoint of smooth operation and product quality control. The contribution of this paper is twofold: firstly, it describes the modelling and simulation aspects for prediction of crystal shape distribution in the presence of crystal growth modifiers (CGMs). Secondly, a control setup is proposed that manipulates the CGM concentration in the crystallizer in order to achieve the desired crystal shape. The effect of CGM on crystal shape is modelled using a morphological population balance model (PBM) in combination with Kubota–Mullin's model for the pinning mechanism of CGM molecules on the crystal face to take into account their possible hindering effect on crystal growth (Kubota and Mullin, 1995; Kubota, 2001). Subsequently, a simple feedback control configuration is implemented in the proposed control setup which manipulates the CGM concentration profile in the crystallizer. The control setup uses a hybrid continuous-batch operation of the crystallizer, in which the solid phase is operated in a batch mode, whereas the liquid phase is operated in a continuous mode maintaining a constant volume in the reactor. This hybrid operation allows the dynamic control of the CGM concentration during the batch. Using potassium dihydrogen phosphate (KDP) as a model system, the simulation results demonstrate that, in principle, the crystal shape distribution can be controlled by manipulating the CGM concentration profile.

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1. Introduction

Crystal size and shape distributions play a crucial role in determining the quality of the final products as well as the efficiency of downstream processes such as filtration. As pointed out by Lovette et al. (2008), the desired crystal size and shape distributions largely depend on its application, *e.g.*, needle-shaped (*i.e.*, high-aspect-ratio) crystals are usually undesirable due to difficulty in processing (Variankaval et al., 2008), while they are vital in applications such as synthesis of ZnO nanowires for solar cells (Law et al., 2005). Moreover, some properties of the crystals such as catalytic activity, dissolution rate, hydrophobicity can be specific to certain crystal face (Snyder et al., 2007). Hence, prediction and control of crystal shape distribution are necessary for product design, quality control and economical operation point of view. This will further enable to develop and adopt the recent trend of Quality-by-Design (QbD) approaches as opposed to conventional Quality-by-Testing (QbT) techniques for designing products with desirable properties (Nagy, 2009; Saleemi et al., 2012; Nagy and Braatz, 2012).

Morphological population balance modelling (PBM), which is multi-dimensional in nature, can be used as a tool to describe the

* Corresponding author at: School of Chemical Engineering, Purdue University, West Lafayette, IN 47907-2100, USA. Tel.: +1 765 494 0734.
E-mail address: zknagy@purdue.edu (Z.K. Nagy).

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evolution of crystal shape distribution in a crystallizer (Gunawan et al., 2004; Ma et al., 2008; Borchert et al., 2009; Majumder et al., 2012). Despite the fact that the effect of impurities (often called additives or crystal growth/nucleation modifiers) on the crystal shape has been demonstrated in many experimental studies (Patience and Rawlings, 2001; Prasad et al., 2001; Kubota et al., 2004; Sangwal, 2007), most of the previous works related to the modelling of crystal shape distributions consider only the effect of operating conditions such as supersaturation (Yang et al., 2006; Lovette and Doherty, 2012), seed loading and cooling rate (Liu et al., 2010; Ma and Wang, 2012), cycles of growth and dissolution (Snyder et al., 2007) and effect of solvents (Davey et al., 1982; Biswas et al., 2007; Liu et al., 2007; Kuvadia and Doherty, 2011) on the crystal shape distribution. The crystal growth modifiers (CGMs) in the form of impurities are often inevitable in industrial crystallization resulting in the inhibition of crystal growth. This growth inhibition further leads to increase in crystallization duration affecting the yield of the process. In addition, selected additives as growth modifiers are used in the industries to achieve the desired crystal shape distribution. In either case, the effect of CGM has to be incorporated in the crystallization modelling to understand the process behavior under realistic operating conditions, when the crystallization occurs in the presence of other compounds in the system.

Recently the effect of CGM has been incorporated in a single dimensional PBM model, to explain the effect of CGMs on the overall growth of particles and on the crystal size distribution (CSD) (Févotte and Févotte, 2010). However, to the best of our knowledge, no work has been presented in the literature, which incorporates the effect of CGM on the evolution of crystal shape distribution. In this work, we present a modelling framework and simulation results that can be used to predict the evolution of crystal shape distribution in a crystallizer in the presence of CGM and at the same time illustrate how the concentration of CGM can be manipulated in a feedback control approach to achieve the desired crystal shape. Additionally the model presented also incorporates a mass balance on the GCM allowing to predict the purity of the final crystals. The growth rate of the crystals depends not only on the concentration of the CGM, but also the time during which the crystals are exposed to the CGM in the mother liquor. These two aspects have recently been incorporated in a mono-dimensional PBM proposed by Févotte and Févotte (2010). We further extend this model to a multi-dimensional PBM so that the resulting morphological PBM is able to describe the evolution of crystal shape distribution in the presence of CGM.

Following the model by Kubota and Mullin (1995) and Kubota (2001), the growth rate dependence on the CGM concentration is described by surface coverage of the adsorbed species on the growing crystal and a parameter denoting the effectiveness of the adsorbed species. It is also assumed that the transient coverage process can be described by a first-order kinetics (Kubota, 2001). On the other hand, in order to track the contact time of the crystals with CGM, the density function is redefined with an additional property. The resulting population balance equations (PBEs) are solved using a well-established semidiscrete finite volume scheme with van Leer flux limiter (van Leer, 1974; LeVeque, 2002) in order to compute the crystal shape distribution. Subsequently, a control setup is proposed for cooling crystallization of potassium dihydrogen phosphate (KDP) using a combined continuous-batch operation. The crystals in the solid phase are operated in a batch mode, whereas the liquid phase is operated continuously by having two inlet flows of CGM and pure solutions, respectively and a corresponding outlet flow of the solid-free mother liquor to maintain constant volume. In this setup, the CGM concentration is manipulated in the crystallizer through a simple feedback control configuration. A Simulink model for the control setup is developed in Matlab where the method of moments (MOM) (McGraw, 1997) using the mixed moments of the bivariate PBM is applied to obtain the average aspect ratio at each sample time. This information is then used by the controller to manipulate the CGM flow rate to control CGM concentration in the crystallizer. With this simple control setup, it is found that the crystal shape distribution can be controlled, even under realistic sampling time.

The paper is organized as follows: the modified crystal growth rate and corresponding PBEs in the presence of CGM are discussed in Section 2. The simulation results demonstrating the effect of CGM, the control setup and aspect ratio control studies are discussed in Section 3 and finally concluding remarks are presented in Section 4.

2. Population balance model for shape distribution in the presence of crystal growth modifiers

In this section, the growth rate model of the crystals in the presence of CGM is discussed briefly. The PBM that takes into account the modified growth rate is also presented.

2.1. Growth rate in the presence of CGM

The growth rate of crystals G_{0i} along the characteristic dimension i in pure solution can be expressed by the following empirical formula (Mullin et al., 1970):

$$G_{0i}(t) = k_{gi}\sigma^{g_i}, \tag{1}$$

$$\sigma = \frac{c - c_{\mathrm{sat}}}{c_{\mathrm{sat}}}, \tag{2}$$

where k_{gi} is the growth parameter, σ is the relative supersaturation, c is the solution concentration, c_{sat} is the saturation concentration and g_i is the growth exponent. The growth of imperfect smooth faces of the crystals takes place by incorporation of the growth units on the kinks of steps (Sangwal, 2007). However, if CGM or impurities are present in the solution, they can get adsorbed on the kinks of the steps, as shown in Fig. 1. As a result, step displacement is pinned at the location of the adsorbed impurity and the step is forced to curve (Kubota, 2001). The step can only move if the radius of the curved step is greater than the radius of the two-dimensional nucleus. The details of this growth model, known as Kubota–Mullin model, can be found in Kubota and Mullin (1995) and Kubota (2001). According to this model, the modified growth rate G_i in the presence of impurity can be expressed as

$$G_i = G_{0i}(1 - \alpha_i\theta_i), \tag{3}$$

$$\alpha_i = \frac{\gamma_i a}{kT\sigma L_i}, \tag{4}$$

where α_i is the effectiveness factor on the ith crystal face accounting for the effectiveness of the CGM under the given

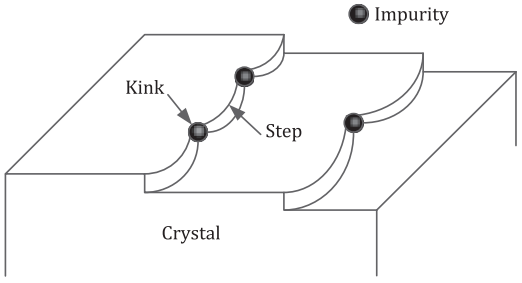


Fig. 1. Impurity adsorption at the surface kinks.

growth condition (*e.g.*, supersaturation, temperature), θ_i is the coverage of the active sites by CGM, γ_i is the edge free energy per unit length of the step on crystal face, a is the surface area of the crystal per growth unit, k is the Boltzmann constant, T is the temperature and L_i is the average distance between the active sites. The dependence of the coverage θ_i on CGM concentration can be expressed by any appropriate isotherm. In many cases the adsorption process proceeds slowly. Thus, kinetics of the adsorption process has to be taken into account as well. In other words, the adsorption process not only depends on the CGM concentration, but also on the time during which the crystals are in contact with CGM. The coverage θ can be expressed as follows for Langmuir adsorption with first-order kinetics (Kubota, 2001):

$$\theta_i = \theta_{\mathrm{eq},i}\left[1-\exp\left(-\frac{\eta}{\tau_i}\right)\right], \tag{5}$$

$$\theta_{\mathrm{eq},i} = \frac{K_i C_{\mathrm{CGM}}}{1+K_i C_{\mathrm{CGM}}}, \tag{6}$$

where $\theta_{\mathrm{eq},i}$ is the equilibrium coverage of the ith crystal face, K_i is the Langmuir constant, C_{CGM} is the concentration of the CGM, η is the time spent by the crystals in contact with CGM and τ_i is the time constant. The contact time η for the crystals depends on the moment they are introduced to the crystallizer. For example, seed crystals introduced at the beginning of the batch will have η same as the batch time, while the newly born crystals due to the secondary nucleation will have η as the time spent starting from the birth time.

It is to be noted that Eq. (3) is valid until the condition $\alpha_i\theta_i < 1$ is satisfied. A characteristic time t_c is defined as follows when crystals cease to grow with $G_i/G_{0i} = 1-\alpha_i\theta_i = 0$:

$$t_c = \tau_i \ln\left(\frac{\alpha_i\theta_{\mathrm{eq},i}}{\alpha_i\theta_{\mathrm{eq},i}-1}\right). \tag{7}$$

Thus, in the presence of CGM that acts as a growth inhibitor, the crystals will continue to grow until the characteristic time t_c is reached.

2.2. Modified multi-dimensional population balance equation for shape distribution in the presence of CGM

Population balance equations (PBEs) are widely used in order to describe the evolution of the crystal size and shape distributions. The dimension of the PBEs depends on the number of the characteristic properties required to describe the size and shape of the crystal. The multi-dimensional PBEs for crystallization processes with growth and nucleation can be written as

$$\frac{\partial}{\partial t}n(t,\mathbf{x}) + \nabla_{\mathbf{x}}\cdot(\mathbf{G}n(t,\mathbf{x})) = B_0\delta(\mathbf{x}-\mathbf{x}_0), \tag{8}$$

$$\text{I.C.}: \; n(0,\mathbf{x}) = n_0(\mathbf{x}), \tag{9}$$

where $n(t,\mathbf{x})$ is the distribution function, $\mathbf{x}=\{x_1,x_2,\ldots,x_D\}$ is the vector of characteristic lengths, $\mathbf{G}=\{G_1,G_2,\ldots,G_D\}$ is the vector of growth rates, B_0 is the rate of nucleation, $\mathbf{x}_0$ is the size of the nuclei, δ is the delta distribution and $n_0(\mathbf{x})$ is the initial distribution. In order to complete the description of the crystallization process, Eq. (8) has to be supplemented by the appropriate mass and energy balance equations.

As discussed in the previous section, in the presence of the CGM it is necessary to keep track of the contact time (η) of the crystals in order to determine the growth rate. Hence, a new distribution function $\phi(t,\mathbf{x},\eta)$ is introduced as follows to incorporate the new property:

$$n(t,\mathbf{x}) = \int_0^t \phi(t,\mathbf{x},\eta)\,d\eta. \tag{10}$$

Then the modified population balance equation describing the evolution of the crystal shape distribution can be written as follows:

$$\frac{\partial}{\partial t}\phi(t,\mathbf{x},\eta) + \nabla_{\mathbf{x}}\cdot(\mathbf{G}\phi(t,\mathbf{x},\eta)) + \frac{\partial}{\partial\eta}\phi(t,\mathbf{x},\eta) = B_0\delta(\mathbf{x}-\mathbf{x}_0)\delta(\eta-0), \tag{11}$$

$$\text{I.C.}: \; \phi(0,\mathbf{x},0) = \phi_0(\mathbf{x}), \tag{12}$$

$$\text{B.C.}: \; \mathbf{G}\phi(t,\mathbf{x},\eta) = 0, \quad \mathbf{x}\in\partial\Omega_{\mathbf{x}}, \;\; \eta = 0, \tag{13}$$

here $\partial\Omega_{\mathbf{x}}$ denotes the boundary of the size space. Note that nucleation is taken into account as a source term on the right side of Eq. (11). One may also implement nucleation as boundary condition since size of the nuclei is often assumed to be zero. There is a range of techniques to solve Eq. (11) for CSD such as finite volume (FV) method with flux limiter also known as high resolution technique (Gunawan et al., 2004; Qamar et al., 2006; Majumder et al., 2010), finite element method (FEM) (Mahoney and Ramkrishna, 2002; Ganesan and Tobiska, 2012), and lattice Boltzmann method (LBM) (Majumder et al., 2012). A semidiscrete implementation of FV method with van Leer flux limiter is used in this paper for its robustness and ease of integration in the Simulink environment provided by Matlab for control studies.

3. Results and discussion

3.1. Process simulation

The cooling crystallization of potassium dihydrogen phosphate (KDP) from aqueous solution is chosen as the system for our simulation studies. This is a well-studied system where the effect of CGM has been investigated experimentally, see *e.g.*, Owczarek and Sangwal (1990a,b), Kubota et al. (2004), Maeda et al. (2006), and Ding et al. (2010). The morphology and molecular arrangement of the KDP crystal are shown in Fig. 2. From this figure it is clear that two characteristic lengths of the KDP crystal are needed to describe the size and shape of the crystal properly. The two families of faces that are exposed to the solution for growth are the prismatic {100} and pyramidal {101} faces. Due to the internal structure of these faces, phosphate ions are exposed to the {100} faces, which therefore exhibit negative charge. Hence, trivalent and divalent metal ions preferably adsorb on the {100} faces

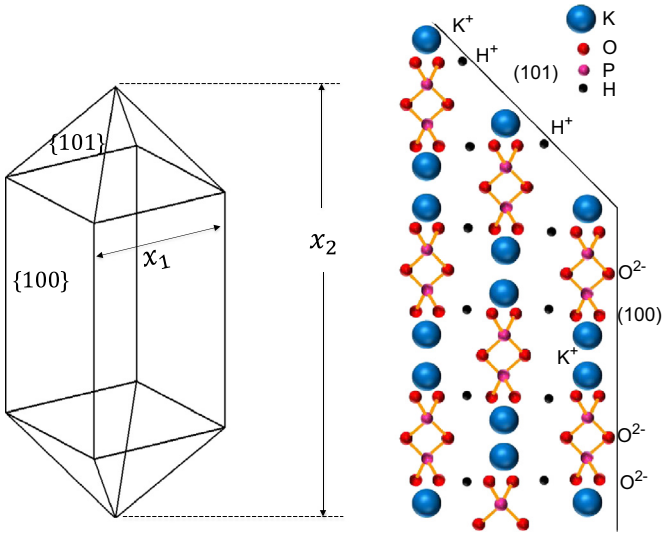


Fig. 2. Morphology and molecular arrangement of the KDP crystal, adapted from Ding et al. (2010).

resulting in the inhibition of the growth step (Rashkovich and Kronsky, 1997). On the other hand, potassium and hydrogen ions are exposed to the {101} faces, which hence exhibit positive charge. As a result, anions with a strong capacity of hydrogen bonding (*e.g.*, oligomeric phosphate ions) can inhibit the growth step of {101} faces (Fu et al., 1999; Ding et al., 2010).

The evolution of the size and shape distributions of the KDP crystal is investigated in two scenarios: one is for pure solution (*e.g.*, without CGM) and the other is when CGM is applied targeting the growth rate in one direction only. The kinetic parameters for crystal growth in pure solution presented in Table 1 are taken from Gunawan et al. (2004). The CGM used in the second scenario is $Al_2(SO_4)_3 \cdot 16.5H_2O$ with Al^{3+} concentration $C_{CGM}=40$ mg/l, which inhibits the growth of {100} faces, *i.e.*, x_1-direction. The required parameters for this study, also shown in Table 1, are taken from Kubota et al. (2004) and Maeda et al. (2006). The initial distribution of the seed crystals are taken as follows with average aspect ratio $r=1.2$:

Table 1
Kinetic parameters for batch crystallization of KDP obtained from Gunawan et al. (2004), Kubota et al. (2004) and Maeda et al. (2006).

Parameter	Value	Units
g_1	1.48	Dimensionless
K_{g1}	12.1	μm/s
g_2	1.74	Dimensionless
K_{g2}	100.75	μm/s
ρ_c	2.338×10^{-12}	g/μm^3
τ_i	36000	s
K_i	0.0648	l/mg
$\frac{\gamma_i a}{kL_i}$	34.15	1/K
$G_{i,min}$	1×10^{-4}	μm/s
$k_{m,o}$	1×10^{-3}	μm/s
K_e	140	Dimensionless

$$n(0,\mathbf{x}) = \begin{cases} -0.00034786(x_1^2 + (x_2-40)^2) + 0.1363609(x_1 + (x_2-40)) \\ -13.2743 \times 2 \quad \text{if } 180 \le x_1 \le 220,\ 220 \le x_2 \le 260 \\ 0 \quad \text{elsewhere \& if } -\text{ve}. \end{cases} \tag{14}$$

The evolution of the size and shape distributions is described by Eq. (11). In these studies the nucleation rate is assumed to be negligible ($B_0=0$), since the primary objective is to investigate the size and shape distributions of the crystals obtained from seed. The mass balance equation for the solute is given as

$$\frac{dc(t)}{dt} = -\rho_c \frac{d}{dt}\left(\int_0^\infty \int_0^\infty \left(x_1^2 x_2 - \frac{2}{3}x_1^3 \tan\theta\right) n(t,\mathbf{x})\, dx_1\, dx_2\right), \tag{15}$$

$$\frac{dc(t)}{dt} = -\rho_c \int_0^\infty \int_0^\infty (2G_1 x_1 x_2 + G_2 x_1^2 - 2G_1 x_1^2 \tan\theta)\, dx_1\, dx_2, \tag{16}$$

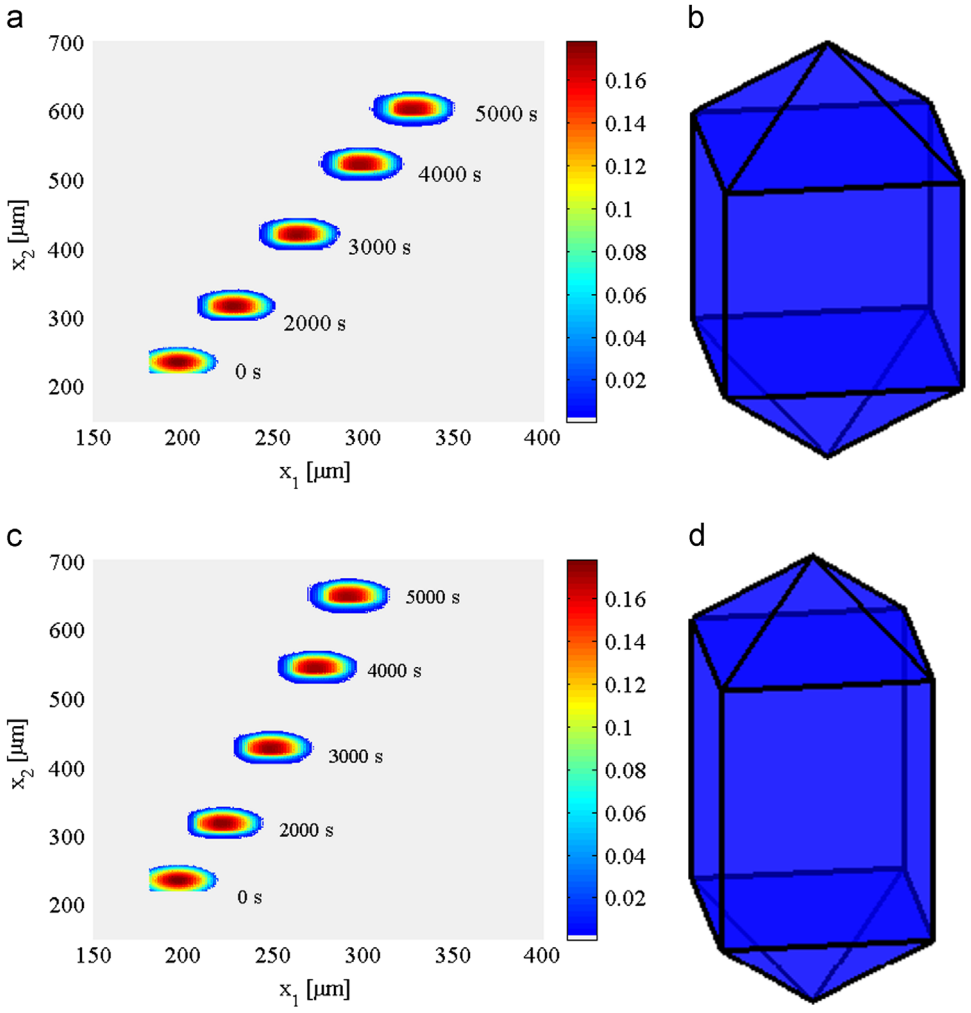


Fig. 3. Effect of CGM on crystal size and shape evolution. Elongated crystals are obtained in the presence of Al^{3+} as CGM. (a) Evolution of crystal size and shape distributions in pure solution. (b) Final crystal shape with $r=1.85$ in pure solution. (c) Evolution of crystal size and shape distributions in the presence of Al^{3+} as CGM (40 mg/l). (d) Final crystal shape with $r=2.23$ in the presence of CGM.

where ρ_c is the crystal density and $v = x_1^2 x_2 - \frac{2}{3}x_1^3 \tan\theta$ is the volume of the crystal. The details on volume calculation for the KDP crystal can be found in the Appendix. The expression for $c_{sat}(T)$ and the temperature profile are given as

$$c_{sat}(T)\ [\text{g/g of water}] = 9.3027 \times 10^{-5} T^2 - 9.7629 \times 10^{-5} T + 0.2087, \tag{17}$$

$$T(t)\ [^\circ\text{C}] = -\frac{33-31}{3600} t + 33. \tag{18}$$

The simulation results for the two scenarios are shown in Fig. 3. The evolution of the crystal size and shape distributions for the pure solution and in the presence of CGM is shown in Fig. 3(a) and (c), respectively. From the evolution of crystal size and shape distributions, it is clear that due to the inhibition of the crystal growth along the x_1-direction in the presence of Al^{3+}, the distance travelled by the distribution along that direction is smaller as compared to the pure solution case. This is also evident from the final shape and average aspect ratio of crystals defined as $r = x_{2,avg}/x_{1,avg}$ shown in Fig. 3(b) and (d) for pure solution and in the presence of Al^{3+}, respectively. Crystals with smaller r are obtained in the first scenario, whereas elongated crystals, *i.e.*, crystals with larger r, are obtained in the second scenario. Thus, the modified growth rate considered in the model is able to describe the effect of the CGM.

3.2. Real-time feedback control of crystal shape using CGM

In order to control the crystal shape by manipulating the CGM concentration in the crystallizer, a simple control setup adapted from Patience and Rawlings (2001) is proposed. As shown in Fig. 4, the two feed streams to the crystallizer are the CGM (F_{CGM} ml/s) and pure solution (F_{in} ml/s) with CGM concentration $C_0 = 250$ mg/l and 0 mg/l, respectively. Solid-free contaminated solution is removed from the crystallizer at the flow rate of F_{out} ml/s to maintain the constant volume of solution. The solute concentration is same in all inlets and the outlet streams. In the simulation it is assumed that the average aspect ratio r is measured in real time. This can be achieved in practice using the appropriate imaging system, either with *in situ* measurement such as the particle vision and measurement probe (PVM) (Braatz, 2002; Kempkes et al., 2008) or on line imaging using a sampling loop with necessary dilution if needed (Schorsch et al., 2012). This measurement is sent to a feedback controller which in turn manipulates the CGM stream to control the CGM concentration in the crystallizer. For example, if the concentration of CGM needs to be increased, the controller increases F_{CGM}. On the other hand, if the CGM concentration needs to be decreased, the controller reduces F_{CGM} and the contaminated solution is flushed by the F_{in} stream. Linear cooling rate is implemented to generate supersaturation. The proposed setup operates in a batch mode from the point of view of crystals, however, it is a continuous system from the point of view of CGM. This setup allows a dynamic manipulation of the CGM concentration (including increase and decrease in its value) during the operation as opposed to the typical batch or semibatch operation mode in which the CGM concentration would accumulate during the batch.

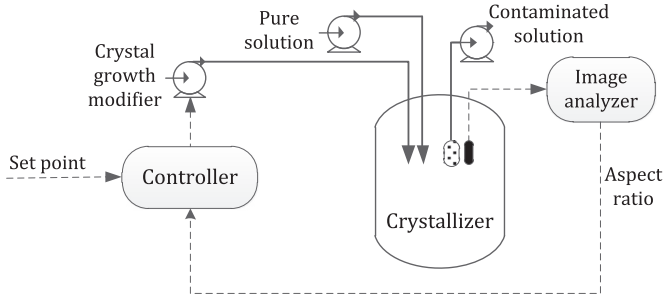


Fig. 4. Schematic of the experimental setup with feedback control.

By doing simple mass and CGM balance around the crystallizer following relations are achieved:

$$F_{out} = F_{CGM} + F_{in}, \tag{19}$$

$$\frac{dC_{CGM}}{dt} = \frac{1}{V}(F_{CGM}C_0 - F_{out}C_{CGM}) + \frac{\chi_c}{1-\chi_c}\frac{MW_{CGM}}{MW_{KDP}}\frac{dc}{dt} \times 10^6, \tag{20}$$

$$\frac{dC_{CGM,c}}{dt} = -\frac{\chi_c}{1-\chi_c}\frac{MW_{CGM}}{MW_{KDP}}\frac{dc}{dt}, \tag{21}$$

where V is the volume of the crystallizer, $C_{CGM,c}$ is the concentration of the CGM in the crystal phase, χ_c is the mole fraction of the CGM in the crystal phase which is related to the mole fraction of the CGM (water free basis) in the solution (χ_{sol}) through the distribution coefficient (K_d), MW_{CGM} and MW_{KDP} are the molecular weights of CGM and KDP, respectively. The first term on the right hand side of Eq. (20) denotes the change in CGM concentration due to inlet and outlet flows, whereas the second term denotes the change in CGM concentration due to adsorption on the crystal face (the constant factor 10^6 appears due to unit conversion). The rate of CGM accumulation on the crystal lattice is described by Eq. (21). The distribution coefficient is defined as

$$K_d = \frac{\chi_c}{\chi_{sol}}. \tag{22}$$

The distribution coefficient can be obtained by the following interfacial distribution model (Maeda et al., 2006):

$$K_d = 1 - (1 - K_e)\left[\frac{G_{i,\min} k_m}{G_i k_{\min}}\right]^{1/2}, \tag{23}$$

$$k_m = \frac{G_i}{1 - \exp(-G_i/k_{m,o})}, \tag{24}$$

$$k_{\min} = \frac{G_{i,\min}}{1 - \exp(-G_{i,\min}/k_{m,o})}, \tag{25}$$

where $G_{i,\min}$ is the specific growth rate when distribution does not occur, k_m and $k_{m,o}$ are mass transfer coefficients on the crystal surface with and without crystal growth, respectively and K_e is the thermodynamic distribution coefficient. The parameters $G_{i,\min}$, $k_{m,o}$ and K_e are obtained by fitting Eqs. (23)–(25) to the experimental data and are shown in Table 1 (Maeda et al., 2006).

The solution of the multi-dimensional PBEs for crystal size and shape distributions is computationally expensive. Since, in the control studies we are interested in controlling the average aspect ratio, r, we can reformulate the PBEs using the method of moments (MOM) to obtain a set of ordinary differential equations (ODEs) and solve it for the moments. For the bivariate PBEs the mixed moments are defined as

$$\mu_{ij} = \int_0^\infty \int_0^\infty n(t, \mathbf{x}) x_1^i x_2^j \, dx_1 \, dx_2. \tag{26}$$

With this definition, the PBE given by Eq. (8) can be written as the following ODE:

$$\frac{d\mu_{ij}}{dt} = iG_1 \langle x_1^{i-1} x_2^j \rangle + jG_2 \langle x_1^i x_2^{j-1} \rangle + B_0 x_{1,0}^i x_{2,0}^j, \tag{27}$$

$$\langle \psi \rangle = \int_0^\infty \int_0^\infty \psi n(t, \mathbf{x}) \, dx_1 \, dx_2, \tag{28}$$

here we are using the notation that for any quantity ψ related to the crystals, the total value of it in the entire size domain is

denoted by $\langle\psi\rangle$. Then the zeroth-, first- and second-order mixed moments can be written as

$$\frac{d\mu_{00}}{dt} = B_0, \tag{29}$$

$$\frac{d\mu_{10}}{dt} = G_1\mu_{00} + B_0 x_{1,0}, \tag{30}$$

$$\frac{d\mu_{01}}{dt} = G_2\mu_{00} + B_0 x_{2,0}, \tag{31}$$

$$\frac{d\mu_{11}}{dt} = G_1\mu_{01} + G_2\mu_{10} + B_0 x_{1,0} x_{2,0}, \tag{32}$$

$$\frac{d\mu_{20}}{dt} = 2G_1\mu_{10} + B_0 x_{1,0}^2, \tag{33}$$

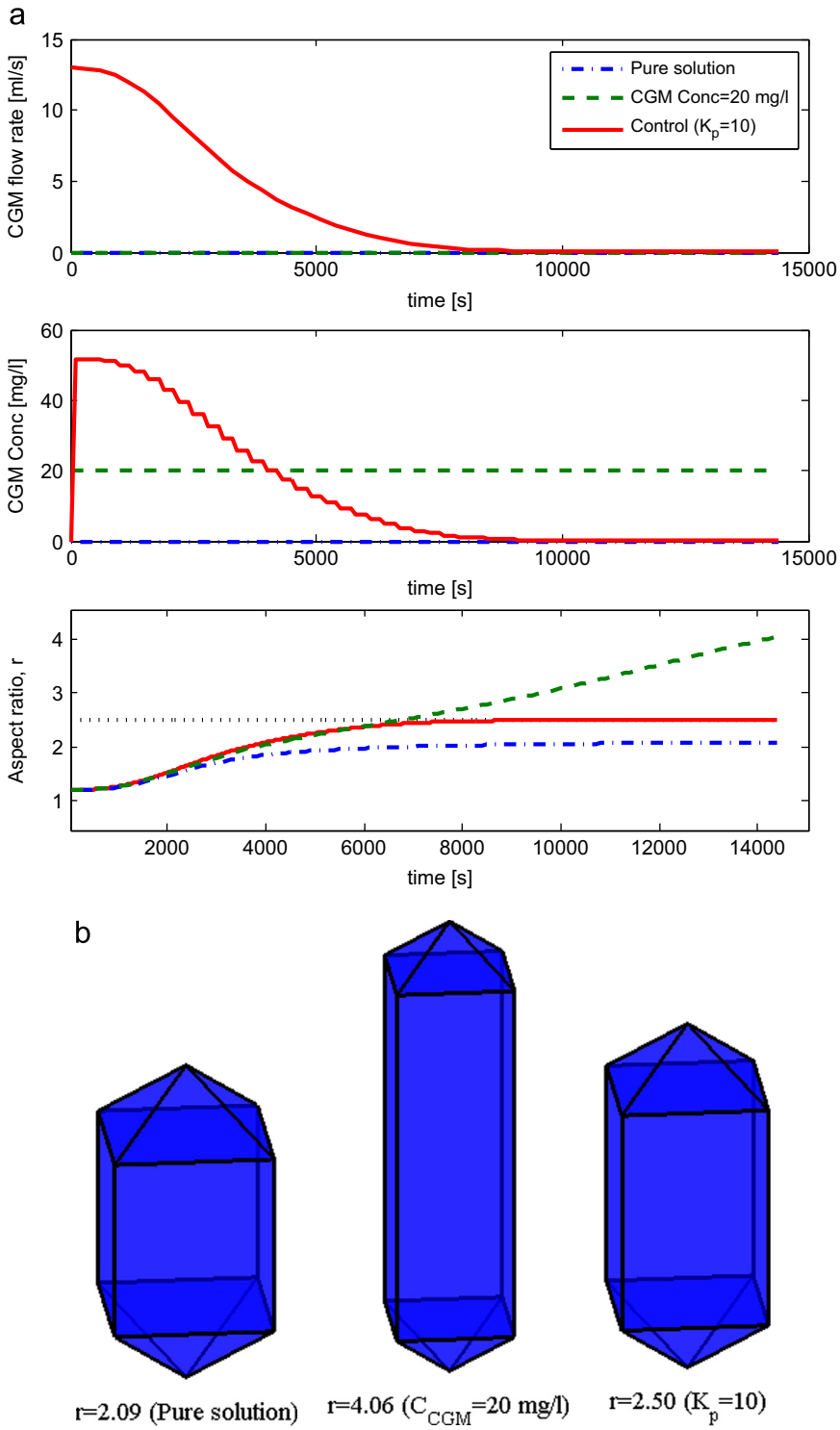


Fig. 5. Control studies when CGM hinders growth in the x_1-direction. A P controller with $K_p=10$ maintains the setpoint of $r_{sp}=2.5$. (a) Flow rate, CGM concentration and aspect ratio profiles. (b) Average aspect ratio (r) of the crystals at the end of the batch.

$$\frac{d\mu_{02}}{dt} = 2G_2\mu_{01} + B_0 x_{2,0}^2. \quad (34)$$

The mass balance equation in terms of moments is written as follows:

$$\frac{dc}{dt} = -\rho_c(2G_1\mu_{11} + G_2\mu_{20} - 2G_1\mu_{20}\tan\theta). \quad (35)$$

Accordingly the average aspect ratio can be defined as

$$r = \frac{\mu_{01}/\mu_{00}}{\mu_{10}/\mu_{00}} = \frac{\mu_{01}}{\mu_{10}}. \quad (36)$$

The set of ODEs obtained in MOM are solved by an explicit fourth-order Runge–Kutta ODE solver implemented in the `ode45` solver in Matlab.

3.3. Aspect ratio control studies

The control setup described in Section 3.2 is implemented in this section, using the crystallizer model to represent the process. We take $V=220$ ml and $F_{in}=50$ ml/s in these studies. The size distribution of the seeds for the control studies is given by Eq. (14). A linear cooling profile given by the following equation is used:

$$T(t)\ [^\circ C] = -\frac{33-23}{3600\times 4}t + 33. \quad (37)$$

In the first study Al^{3+} is used as CGM which affects the growth in the x_1-direction. The profiles for the CGM flow rate F_{CGM}, the CGM concentration in the crystallizer C_{CGM} and the average aspect ratio r are shown in Fig. 5. As can be seen from these plots, when pure solution is used without any control setup, starting with an initial

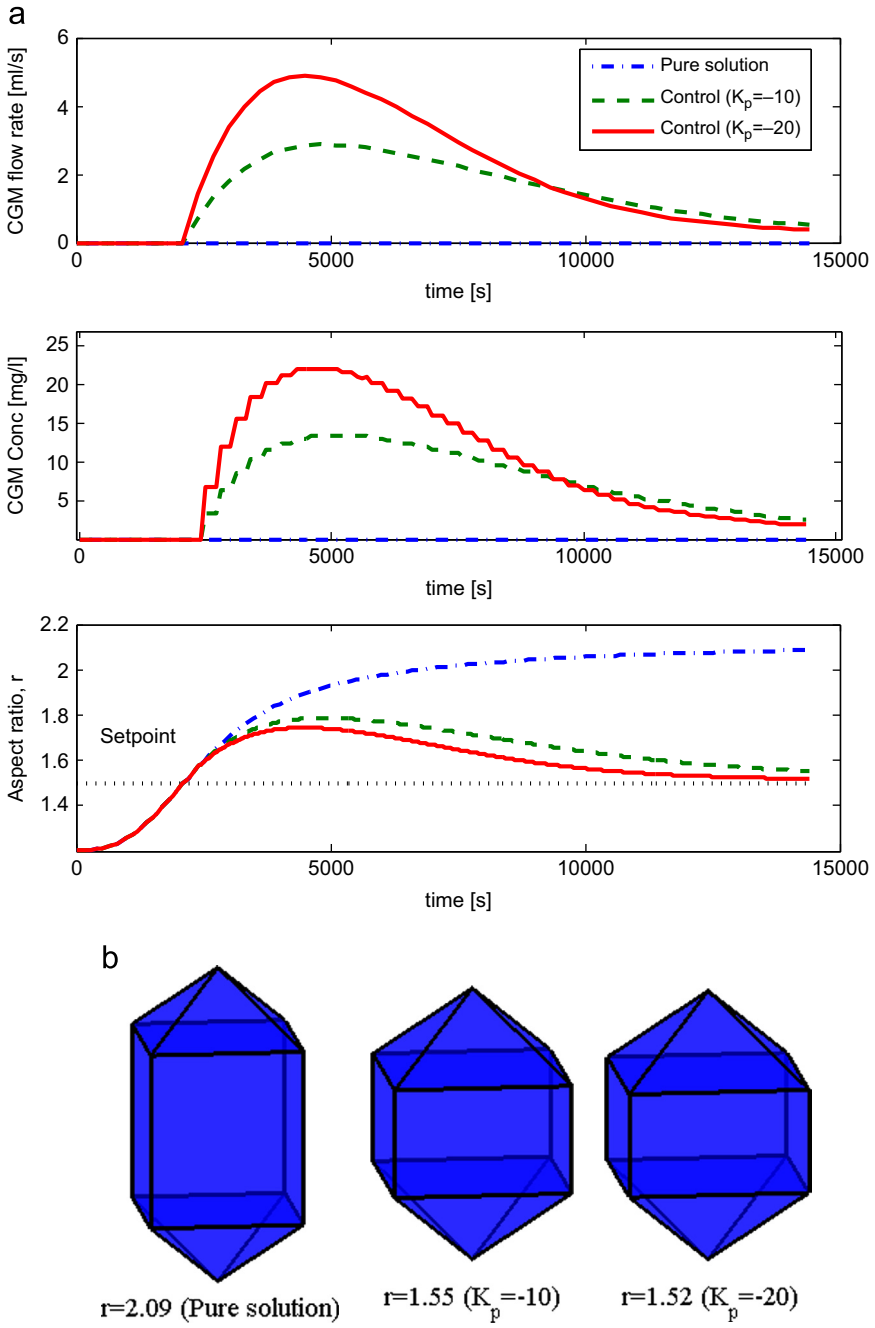


Fig. 6. Control studies when CGM hinders growth in the x_2-direction. A P controller with $K_p=-20$ maintains the setpoint of $r_{sp}=1.5$. (a) Flow rate, CGM concentration and aspect ratio profiles. (b) Average aspect ratio (r) of the crystals at the end of the batch.

average aspect ratio of the seed crystals of $r=1.2$, the final value of $r=2.09$ is obtained after a batch time of 14,400 s (4 h). On the other hand, when Al^{3+} is present at an approximately constant concentration of $C_{CGM}=20$ mg/l throughout the batch processes, very elongated crystals with $r=4.06$ are obtained. Thus, in principle one should be able to manipulate Al^{3+} concentration to obtain crystals with $2.09 < r < 4.06$. Next, the same simulation is performed within the feedback control framework for a setpoint $r_{sp}=2.5$. As can be seen from Fig. 5, a simple proportional controller that takes a control action every 300 s with a proportional gain of $K_p=10$ works well in this case. The 300 s sampling time is chosen since initial experimental investigations indicated that up to 5 min sampling time is needed to acquire enough good quality images to compute a representative mean aspect ratio by the tested imaging systems and image analysis tools. The governing equation of the proportional controller can be written as

$$F_{CGM} = F_{CGM,0} + K_p(r_{sp} - r), \tag{38}$$

where $F_{CGM,0}$ is the initial value of the CGM flow rate. It is found that the setpoint is reached at around $t=6500$ s without any overshoot and the setpoint is maintained for the rest of the simulation. One can see in Fig. 5 that the controller takes the right action by increasing the F_{CGM} in order to increase C_{CGM} in the crystallizer at the initial stage of the crystallization process which in turn increases the aspect ratio. After reaching the setpoint of $r=2.5$, a very low concentration of C_{CGM} is maintained by the controller needed to maintain the setpoint. It is also worth noting that the off set is negligible in this study.

In the next study, the control of the crystal shape is investigated by manipulating CGM that affects the growth of pyramidal {101} face and eventually the growth in the x_2-direction. Although experimental studies show that anions with a strong capacity of hydrogen bonds preferably get adsorbed on the pyramidal face causing the inhibition in growth of that face (Fu et al., 1999), to the best of our knowledge the kinetic parameters for such studies are not available. Thus, in order to perform the simulation of the effect of CGM that affects pyramidal face, similar parameters as Al^{3+} are used. The control studies are shown in Fig. 6. It can be seen that when pure solution is used, crystals with $r=2.09$ are obtained as in the previous case. This time the target is to achieve crystals with smaller aspect ratio and the setpoint $r_{sp}=1.5$ is used. From the plots in Fig. 6(a), it is seen that the controller maintains $C_{CGM}=0$ by setting $F_{CGM}=0$ until the aspect ratio increases from $r=1.2$ to the setpoint $r=1.5$. Subsequently there is an overshoot and the C_{CGM} is increased by manipulating F_{CGM}. Eventually, the target average aspect ratio $r=1.5$ is maintained by keeping a very low concentration of CGM in the crystallizer. Again, in this case a simple P control that takes action every 300 s is found to be working well. A higher proportional gain reduces the offset to a practically negligible level. These results indicate that a simple practical feedback control implementation using a realistic 300 s sampling time for image analysis and a simple proportional feedback control law can provide adequate aspect ratio control by manipulating the CGM flow rate. The proposed setup is able to dynamically manipulate (increase and decrease) the CGM concentration during the batch. While the CGM concentration during the first half of the batch is high, at the end the CGM concentration in the solution is significantly lower.

To investigate the effect of the CGM-based aspect ratio feedback control on the purity of the crystals, the amount of CGM built into the crystal lattice during the batch is computed by integrating the CGM adsorption term in Eq. (21) for the duration of the batch, for all control scenarios as well as for the uncontrolled case. The results are shown in Table 2. It can be seen that very low amounts of CGM can have a significant effect on the crystal aspect ratio.

Table 2
Amount of CGM adsorbed in the crystal lattice.

Case	CGM% in wt.
No control ($C_{CGM}=20$ mg/l)	0.24
Control study 1 ($K_p=10$)	0.15
Control study 2 ($K_p=-10$)	0.12
Control study 2 ($K_p=-20$)	0.16

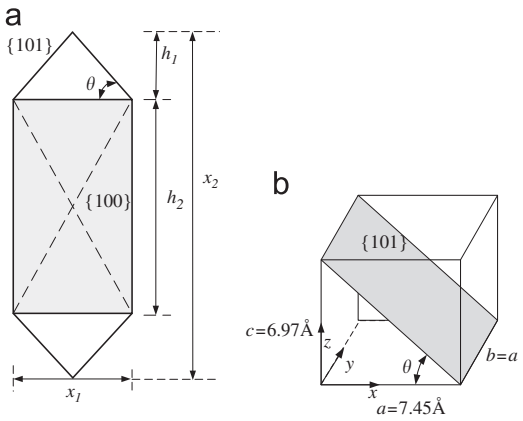


Fig. 7. (a) Schematic of the KDP crystal for volume calculation. (b) Unit cell showing (101) face.

In all cases the final crystal purity is higher than 99.7% (in weight). In the uncontrolled case only 0.24% (in weight) CGM doubles the aspect ratio. When the crystal aspect ratio is controlled at lower target value the final purity is higher. In all controlled cases evaluated in this work the final purity of crystals is above 99.8%. From the controller tuning perspective, the results show that although with a higher gain ($K_p=-20$) the target aspect ratio is achieved earlier during the batch, the purity of the final crystal is slightly lower than for the smaller controller gain ($K_p=-10$). This is due to the fact that the more aggressive control with larger gain creates higher CGM concentration levels during the first half of the process, during which more additive can be incorporated in the crystal lattice. These results indicate that the tuning of the controller will not only influence how well the target aspect ratio will be controlled but can also have an effect on the final crystal purity.

In these studies a simple proportional control is used to provide a proof of concept for a practically easily implementable setup that can be used for crystal shape manipulation using CGM. The proposed setup can also be used in a more complex nonlinear model predictive control framework, in which various operational constraints can also be incorporated (*e.g.*, maximum admissible CGM concentration in the crystal lattice in the final product).

4. Conclusions

The modelling framework presented in this work combines the adsorption model of the crystal growth modifier (CGM) on the crystal surface to describe the growth inhibition with the morphological population balance equations (PBEs) to predict the crystal size and shape distributions, as well as the purity of the crystals. The simulation results confirmed that very small amounts of impurities in the crystal lattice can have a significant effect on the crystal shape. Further, it is shown that in principle one can manipulate the CGM concentration in a simple feedback control

configuration to control the crystal size and shape distributions. Although, a simple proportional controller is found to work well in simulation studies, due to the inherent dynamics of the instruments and equipments such as valves and pumps, a PID controller may be required in real practice. In future experiments will be carried out to validate these simulation results.

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Appendix A. Volume calculation for KDP crystal

The size of the KDP crystal is characterized by x_1 and x_2 lengths in Cartesian coordinates as shown in Fig. 7. The crystallographic information such as the angle θ between the {101} face (pyramidal) and {001} face (prismatic) is required to calculate the volume of the crystal. The unit cell for KDP is body centred tetragonal with sides $a=b=7.45$ Å, $c=6.97$ Å (Nelmes et al., 1987) as shown in Fig. 7(b). Thus, the angle between the {101} face (pyramidal) and {001} face (prismatic) can be calculated as

$$h_1 = \frac{x_1}{2} \tan\theta, \tag{39}$$

$$h_2 = x_2 - x_1 \tan\theta. \tag{40}$$

Now the volume of crystal can be calculated as follows:

$$v = 2 \times \text{volume of the pyramid} + \text{volume of the prism} \tag{41}$$

$$v = 2\left(\tfrac{1}{3}x_1^2 h_1\right) + x_1^2 h_2, \tag{42}$$

$$v = x_1^2 x_2 - \tfrac{2}{3}x_1^3 \tan\theta. \tag{43}$$

Notations

a	area of the crystal per growth unit, μm^2
c	solute concentration, g/g of water
C_{CGM}	concentration of the CGM, g/l
$C_{CGM,c}$	concentration of the CGM in the crystal phase, g/g of water
C_0	concentration of the CGM at the inlet stream, g/l
F_{CGM}	inlet flow rate of feed solution containing CGM, ml/s
F_{in}	inlet flow rate of pure feed solution, ml/s
F_{out}	out flow rate of contaminated solution, ml/s
G_i	growth rate in the ith direction, $\mu m/s$
$G_{i,min}$	growth rate in the ith direction when distribution does not occur, $\mu m/s$
k_m	mass transfer coefficient with crystal growth, $\mu m/s$
$k_{m,o}$	mass transfer coefficient without crystal growth, $\mu m/s$
K_d	effective distribution coefficient, $\mu m/s$
K_e	thermodynamic distribution coefficient, $\mu m/s$
K_i	Langmuir constant, l/gm
K_p	proportional gain, dimensionless
MW_{CGM}	molecular weight of CGM, g
MW_{KDP}	molecular weight of KDP, g
n	size and shape distributions, $\#/\mu m^2$
r	average aspect ratio, dimensionless
t	time, s
v	volume of a crystal, μm^3
V	volume of the crystallizer, μm^3
x_i	characteristic length in the ith direction, μm

Greek letters

α_i	effectiveness factor, dimensionless
γ_i	edge free energy per unit length, J/m
μ_{ij}	mixed moment of distribution, μm^{i+j}
ρ_c	density of the crystal, $gm/\mu m^3$
σ	relative supersaturation, dimensionless
τ_i	adsorption time constant, s
θ_i	adsorption coverage, dimensionless
ϕ	size and shape distributions, $\#/(\mu m^2\ s)$
η	contact time of the crystals, s
χ_c	mole fraction of the CGM in the crystal phase, dimensionless
χ_{sol}	mole fraction of the CGM in the solution (water free basis), dimensionless

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